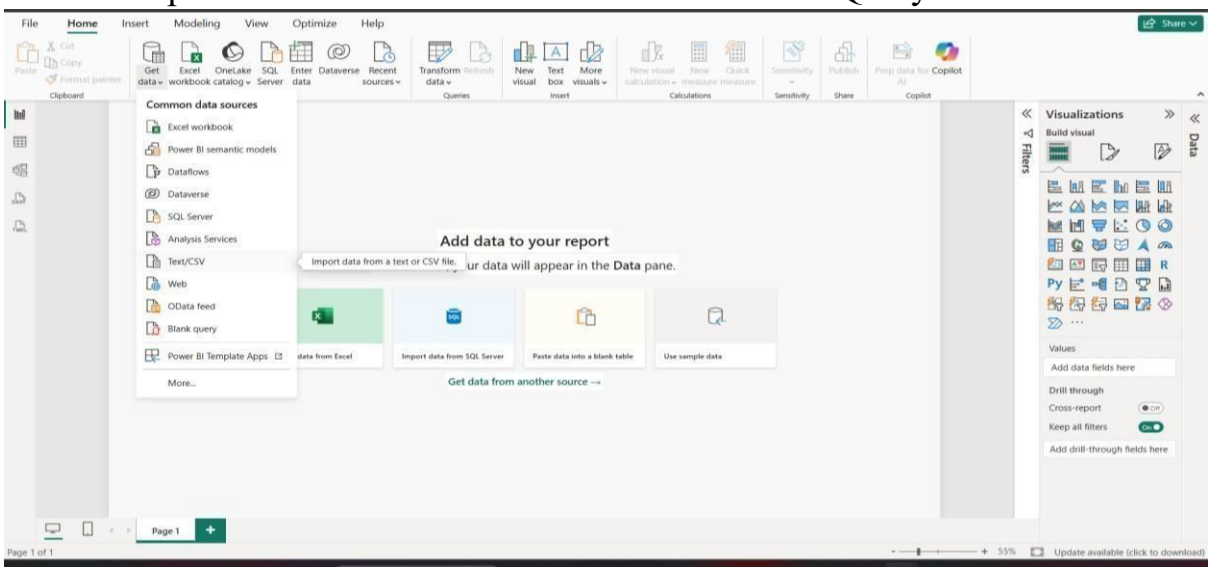
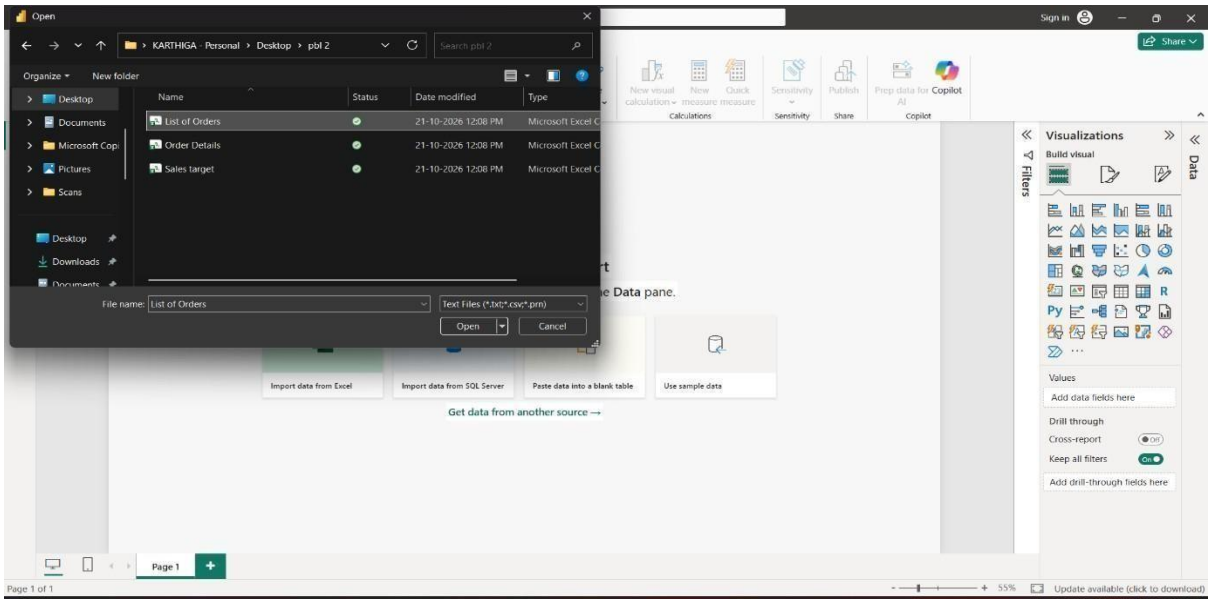
Power BI Assignment 2 – DAX & Data Visualization

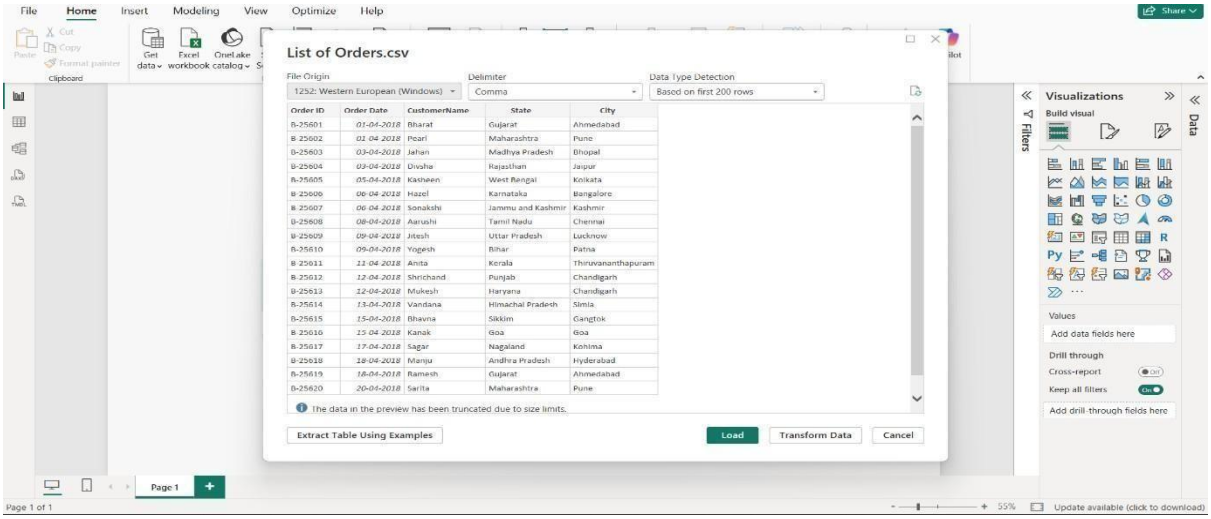
Here , the three E- Commerce Sales Analysis datasets are given CSV format .

- ★ **List of Orders.csv**
- ★ **Order Details.csv**
- ★ **Sales target.csv**

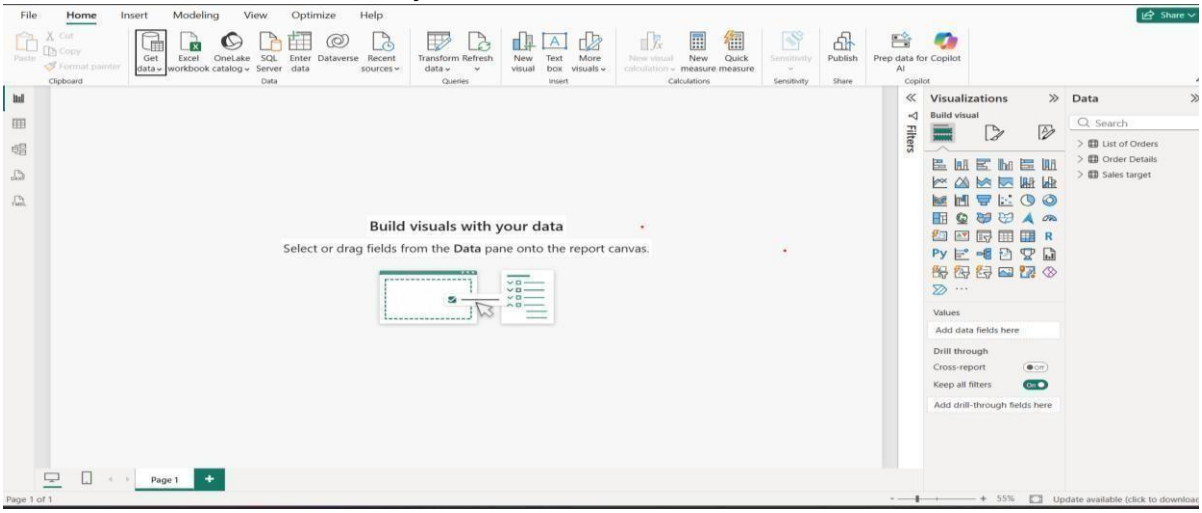
These files are imported into the Power BI and into the Power Query Editor.

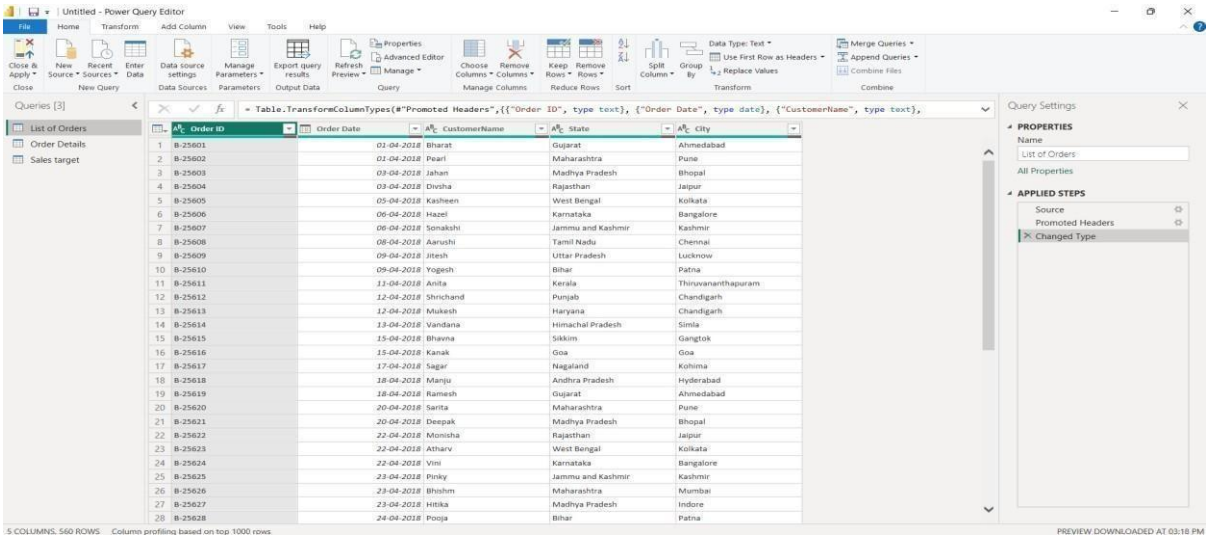




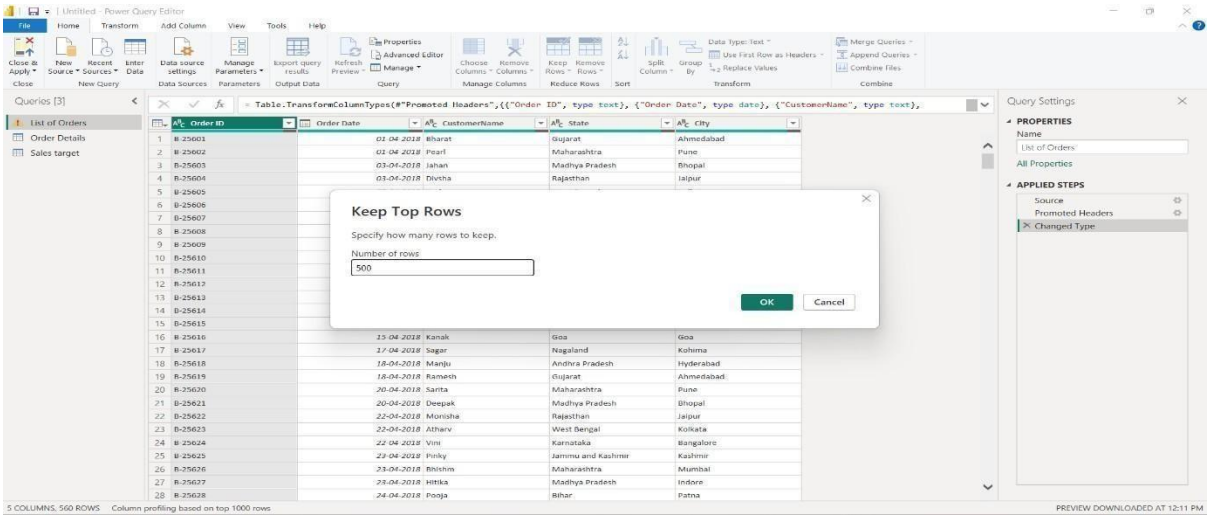


The same procedure is repeated for the remaining 2 CSV files and are imported into the PowerBI and Power Query Editor.

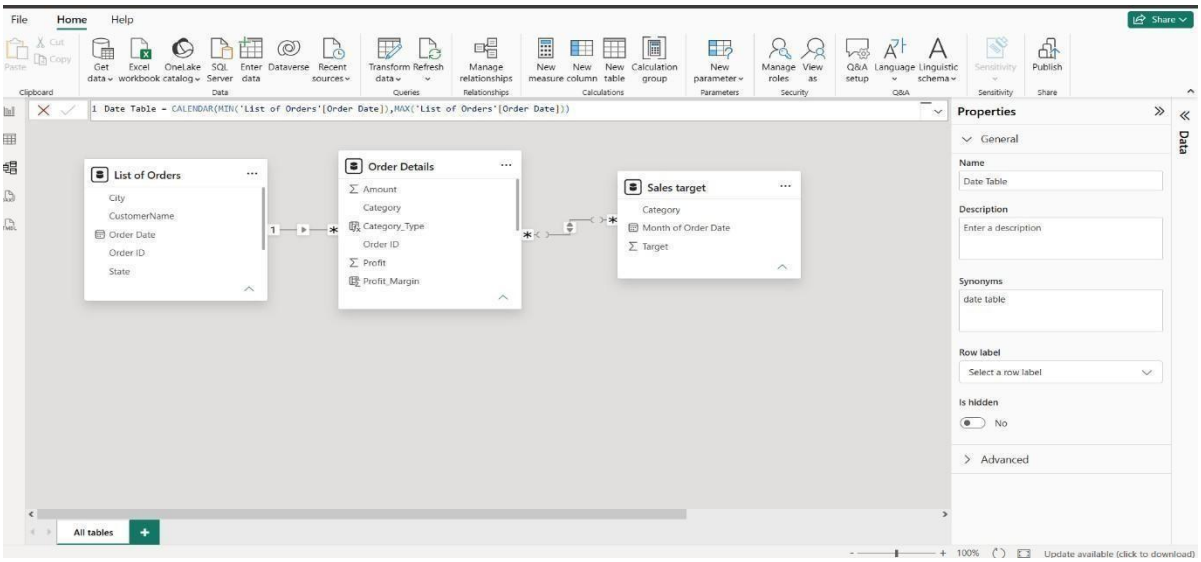




In the LIST OF ORDERS table , there are 560 rows totally with 60 NULL valued rows at the bottom . So, we can remove those NULL valued rows and keep top 500 rows for further Analysis.



DATA MODELING



The Relationship between the given 3 datasets are established.

List Of Orders (Order Id) ----- Order Details (Order Id)

Cardinality – Many to One Cross Filter Directions – Single

Order Details (Category) ----- Sales Target (Category)

Cardinality – Many to Many Cross Filter Directions – Both

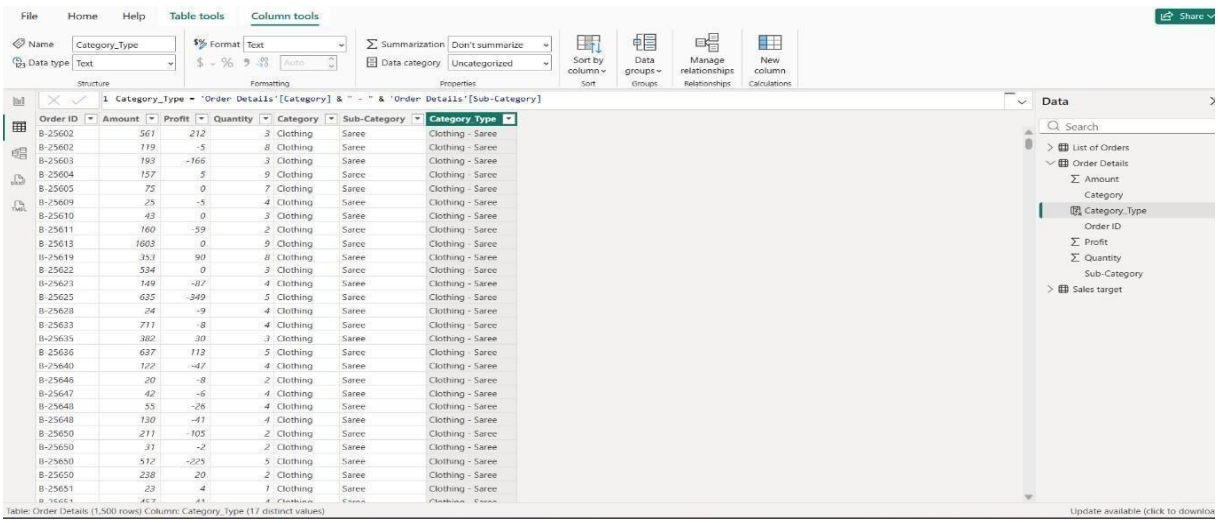
CALCULATED COLUMNS

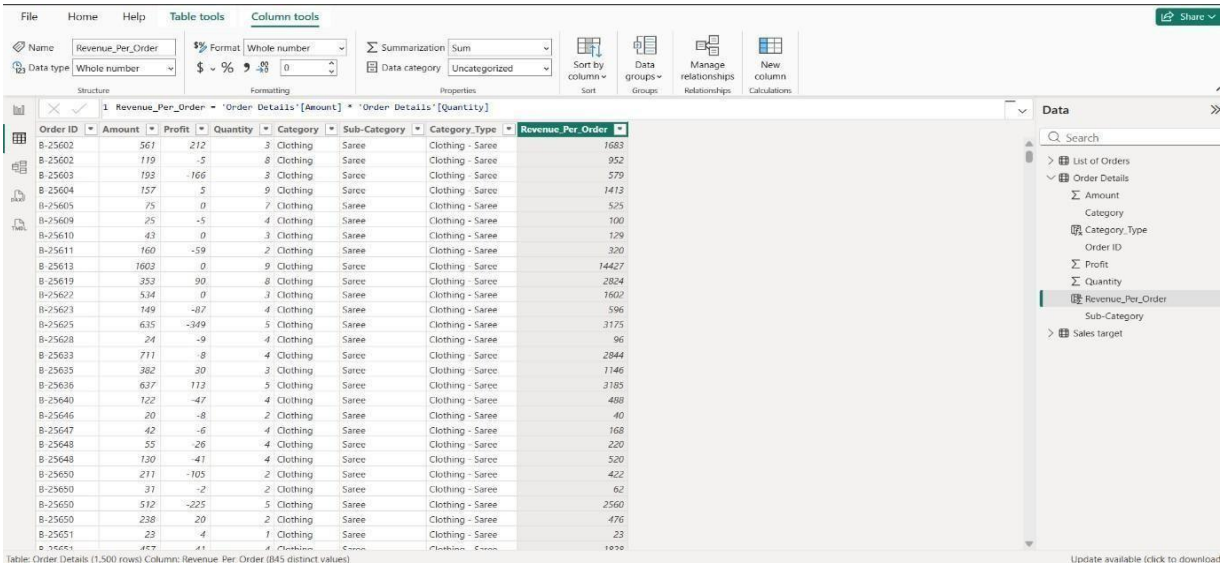
In the ORDER DETAILS table , the 3 columns are created and are named as CATEGORY_TYPE , REVENUE_PER_ORDER , SALES_CATEGORY .

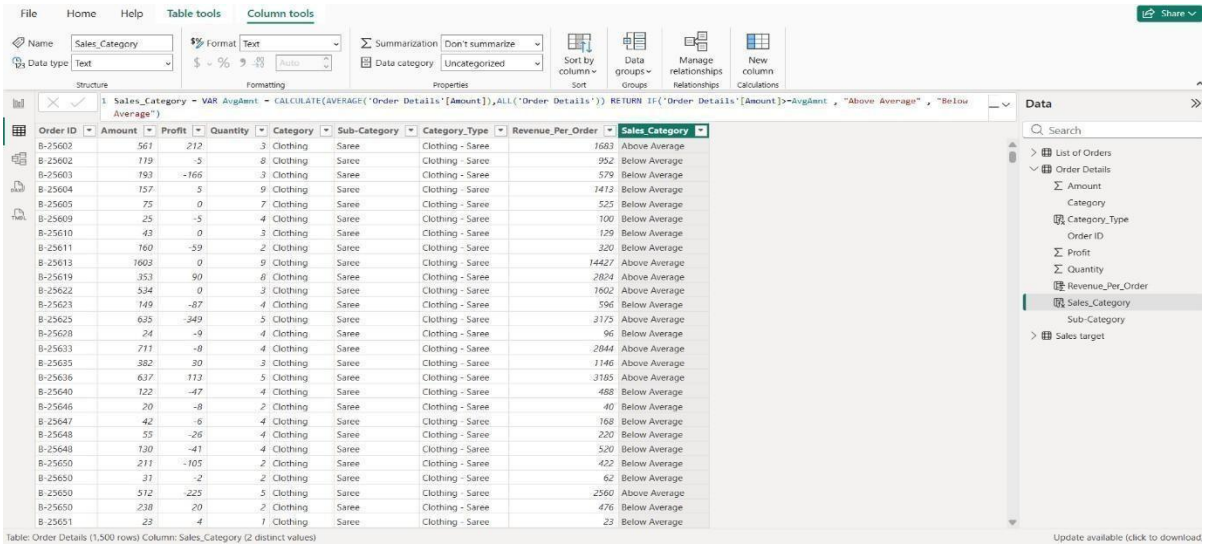
CATEGORY _TYPE = 'Order Details'[Category] & " - " & 'Order Details'[SubCategory]

REVENUE_PER_ORDER = 'Order Details'[Amount] * 'Order Details'[Quantity]

SALES_CATEGORY = VAR AvgAmnt = CALCULATE(AVERAGE('Order Details'[Amount]),ALL('Order Details')) RETURN IF('OrderDetails'[Amount]>=AvgAmnt , "Above Average" , "Below Average")







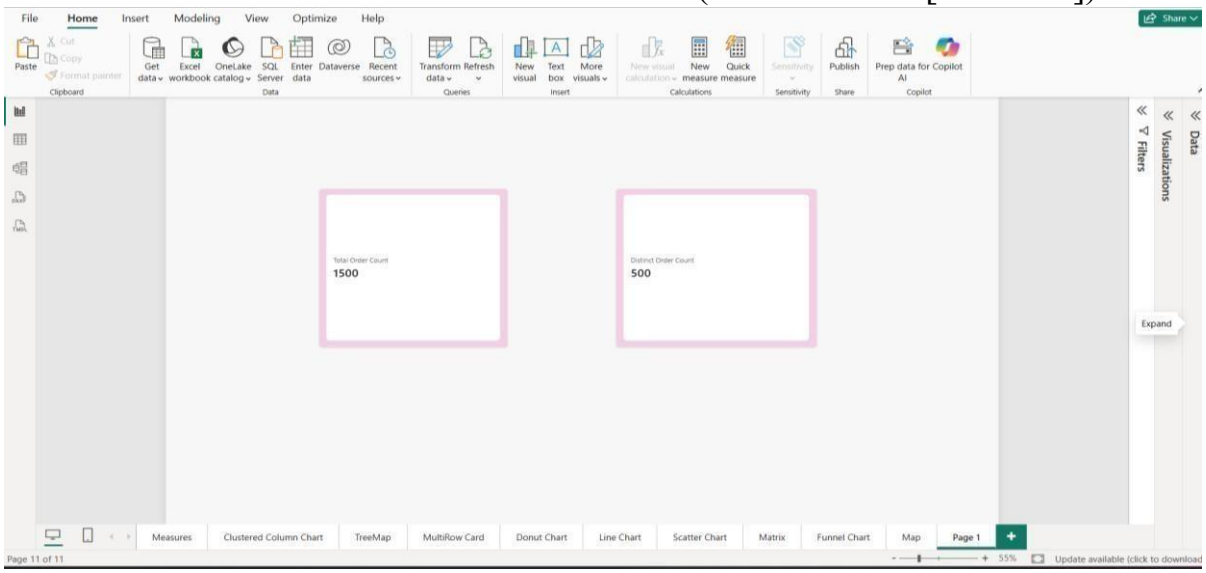
CALCULATED MEASURES

1. CALCULATE ORDER COUNT

In the ORDER DETAILS Table , the COUNTROWS () and DISTINCTCOUNT () functions are used to find the total order count and the distinct order count.

TOTAL ORDER COUNT = COUNTROWS('Order Details')

DISTINCT ORDER COUNT = DISTINCTCOUNT('Order Details'[Order ID])

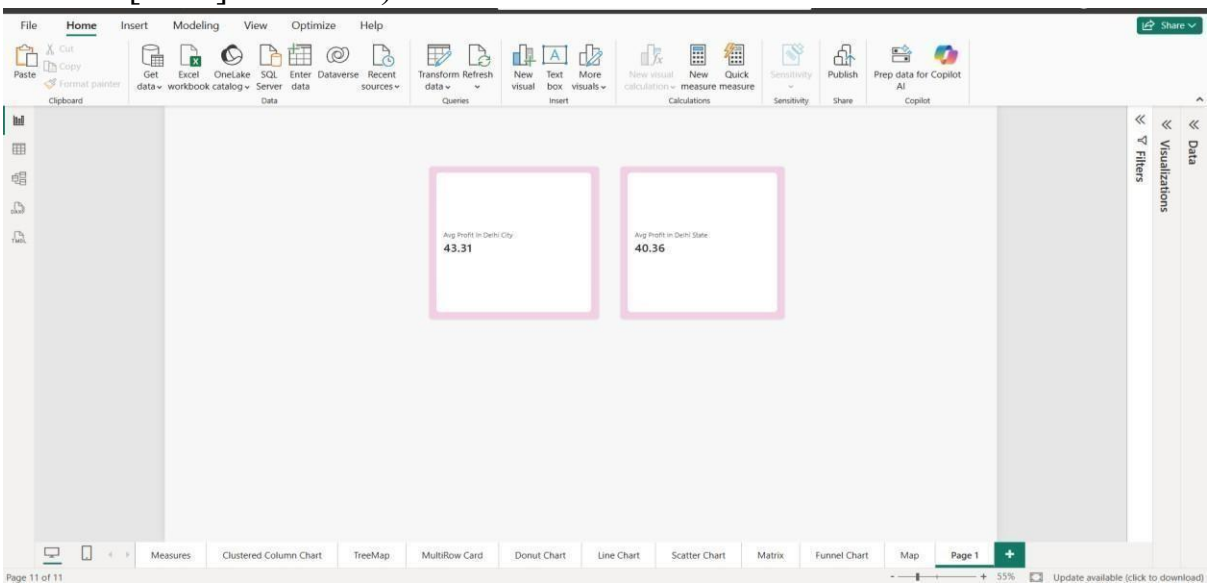


2. CALCULATE AVERAGE PROFIT IN DELHI

In the LIST OF ORDERS Table , DELHI is presented in both CITY and STATE columns. So, 2 measures are calculated.

Avg Profit In Delhi City = CALCULATE(AVERAGE('Order Details'[Profit]) , 'List of Orders'[City] = "Delhi")

Avg Profit In Delhi State = CALCULATE(AVERAGE('Order Details'[Profit]) , 'List of Orders'[State] = "Delhi")



3. CALCULATE YEAR TO DATE (YTD) SALES

Here, the New table named DATE TABLE is created using the NEW TABLE option. This table contains 5 columns named DATE, YEAR, MONTH, MONTH_NAME, MONTH_YEAR.

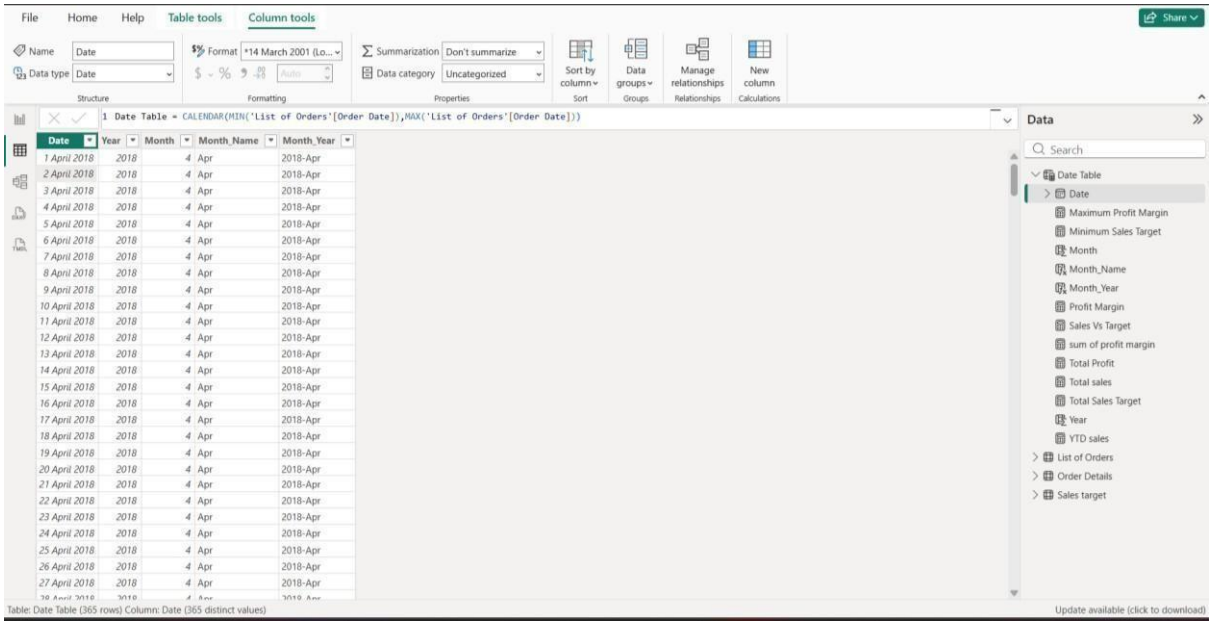
DATE TABLE = CALENDAR(MIN('List of Orders'[Order Date]),MAX('List of Orders'[Order Date]))

MONTH = MONTH('Date Table'[Date])

YEAR = YEAR('Date Table'[Date])

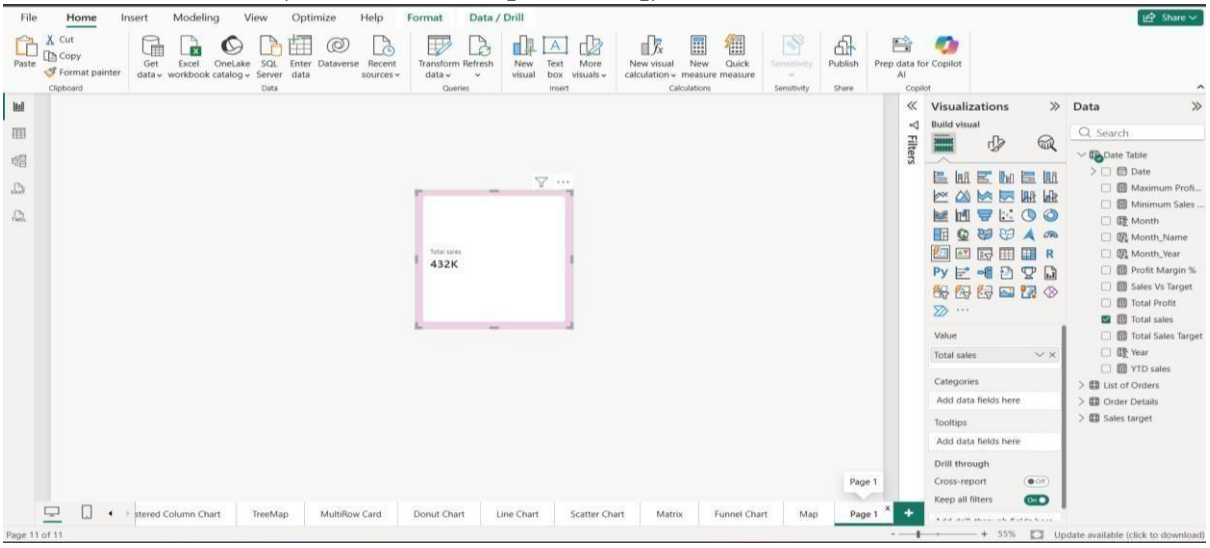
MONTH_NAME = FORMAT('Date Table'[Date], "MMM")

MONTH_YEAR = FORMAT('Date Table'[Date], "YYYY-MMM")



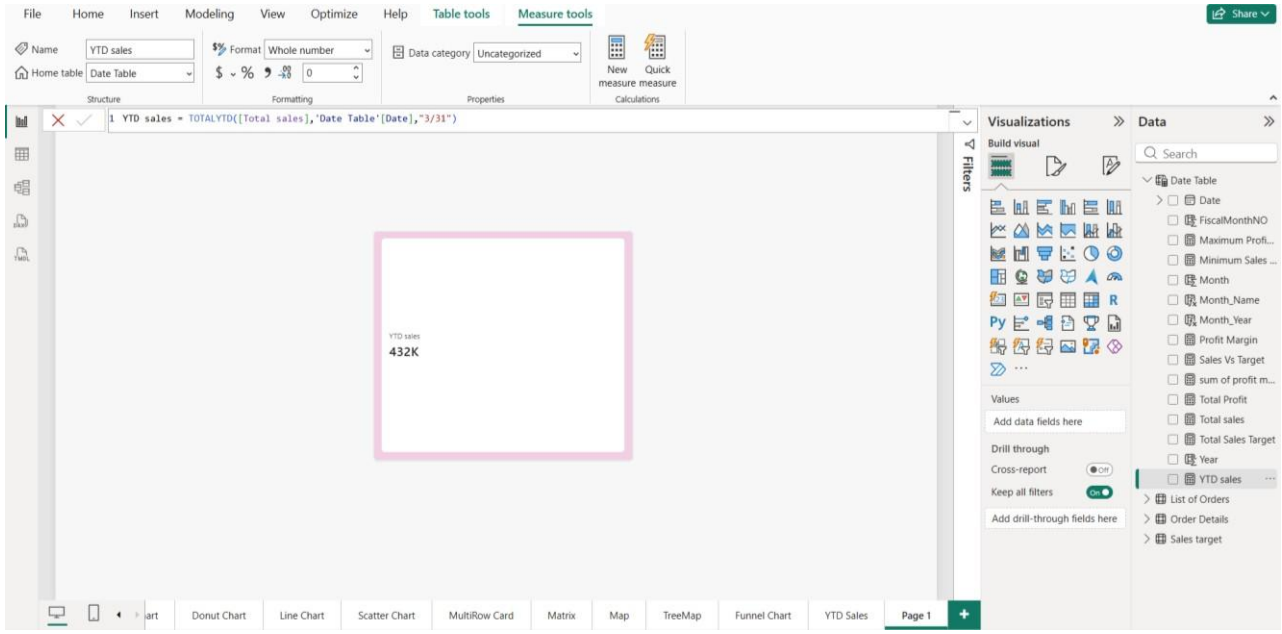
Now, TOTAL SALES is created as a NEW MEASURE.

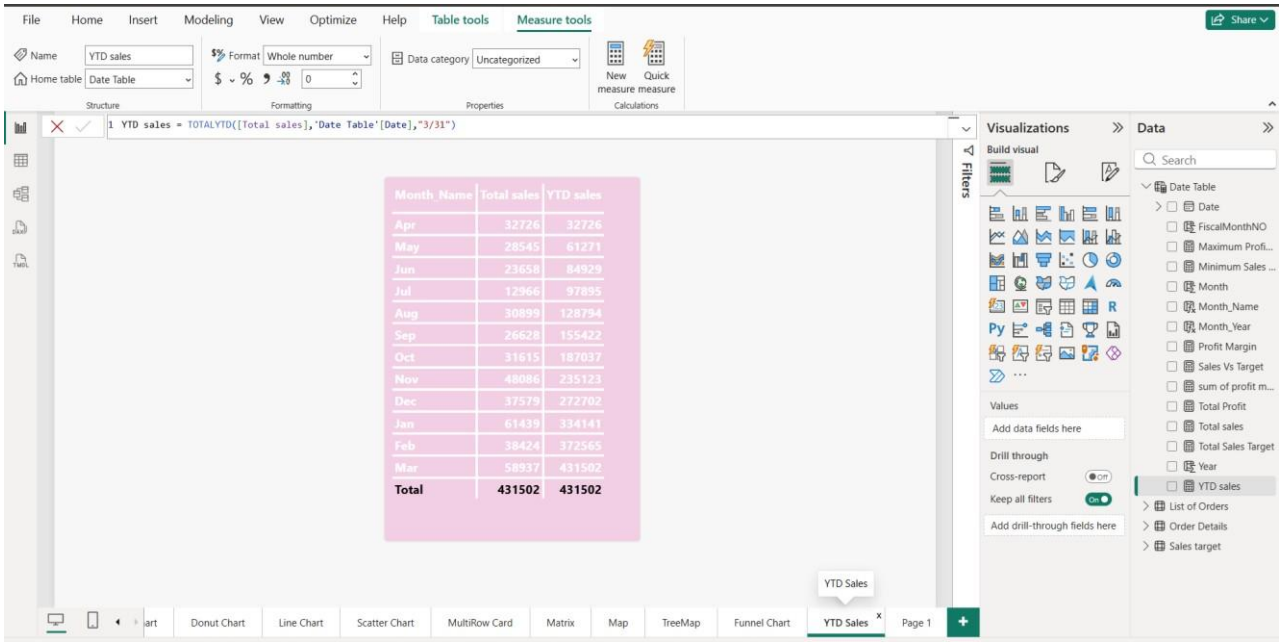
TOTAL SALES = SUM('Order Details'[Amount])



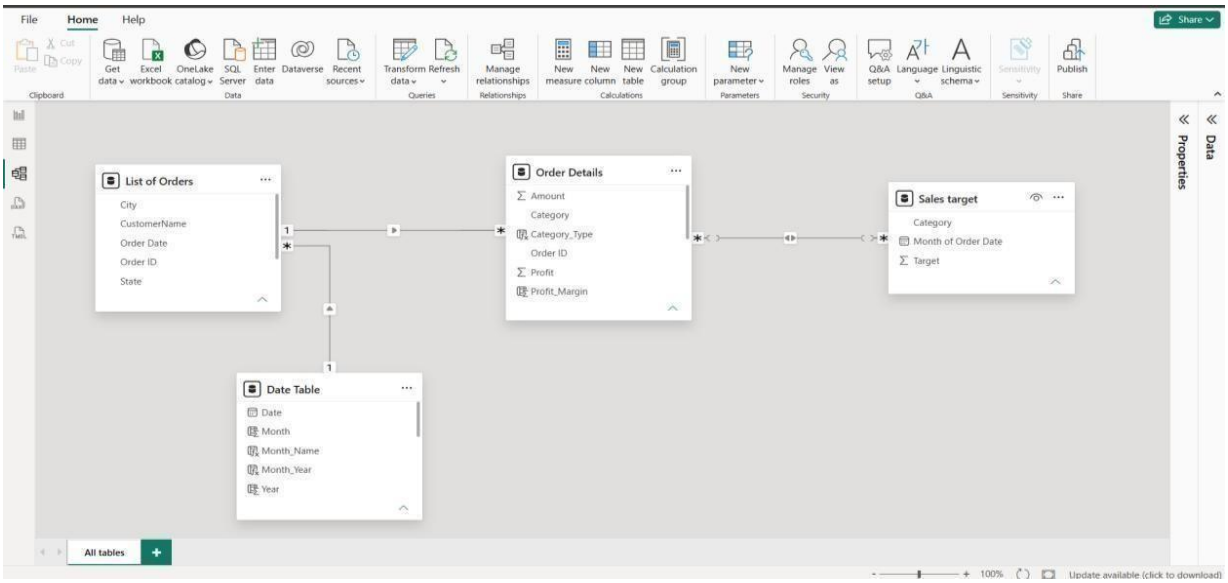
Now , YTD SALES measure is calculated using the formula.

YTD SALES = TOTALYTD([Total sales],'Date Table'[Date],"3/31")


File Home Insert Modeling View Optimize Help Table tools Measure tools
Share
Name YTD sales
Home table Date Table
Structure
Format Whole number
Formatting
Data category Uncategorized
Properties
New measure
Quick measure
Calculations
1 YTD sales = TOTALYTD([Total sales],'Date Table'[Date],"3/31")
YTD sales
432K
Visualizations
Build visual
Values
Add data fields here
Drill through
Cross-report
Keep all filters
Add drill-through fields here
Data
Search
Date Table
Date
FiscalMonthNO
Maximum Profi...
Minimum Sales ...
Month
Month_Name
Month_Year
Profit Margin
Sales Vs Target
sum of profit m...
Total Profit
Total sales
Total Sales Target
Year
YTD sales
List of Orders
Order Details
Sales target
Filters
Donut Chart
Line Chart
Scatter Chart
MultiRow Card
Matrix
Map
TreeMap
Funnel Chart
YTD Sales
Page 1


File Home Insert Modeling View Optimize Help Table tools Measure tools
Share
1 YTD sales = TOTALYTD([Total sales],'Date Table'[Date],"3/31")
Month_Name Total sales YTD sales
Apr 32726 32726
May 28545 61271
Jun 23658 84929
Jul 12966 97895
Aug 30899 128794
Sep 26628 155422
Oct 31615 187037
Nov 48086 235123
Dec 37579 272702
Jan 61439 334141
Feb 38424 372565
Mar 58937 431502
Total 431502 431502
YTD Sales
Visualizations
Data
Donut Chart
Line Chart
Scatter Chart
MultiRow Card
Matrix
Map
TreeMap
Funnel Chart
YTD Sales
Page 1

The Relationship is created between all the 4 tables ., LIST OF ORDERS, ORDER DETAILS, SALES TARGET AND DATE TABLE.



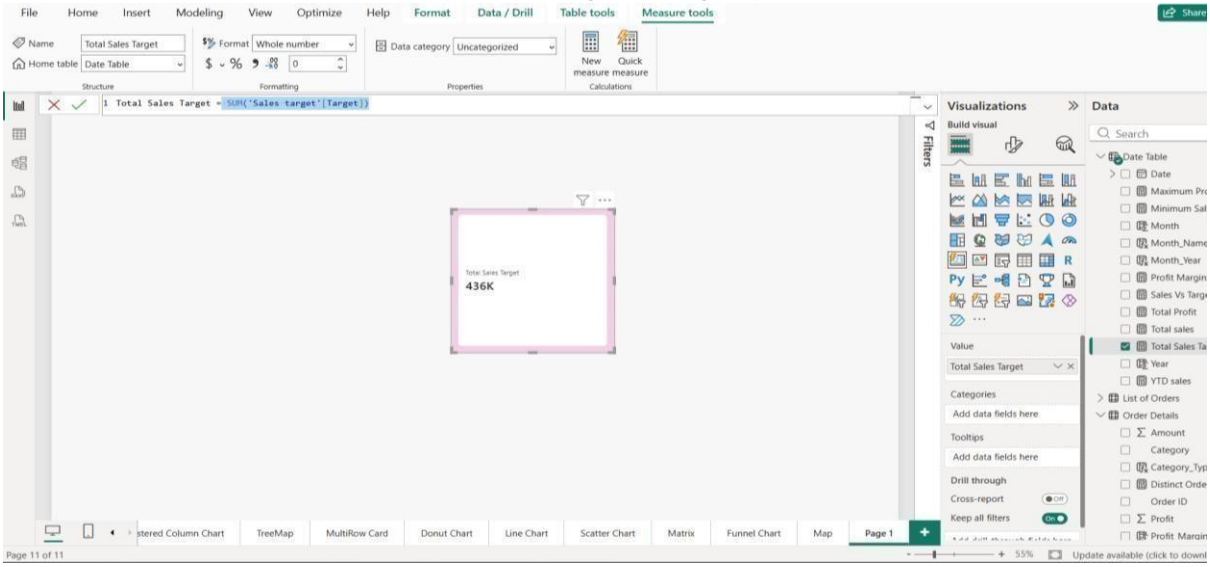
List Of Orders (Order Date) ----- Date Table (Date)

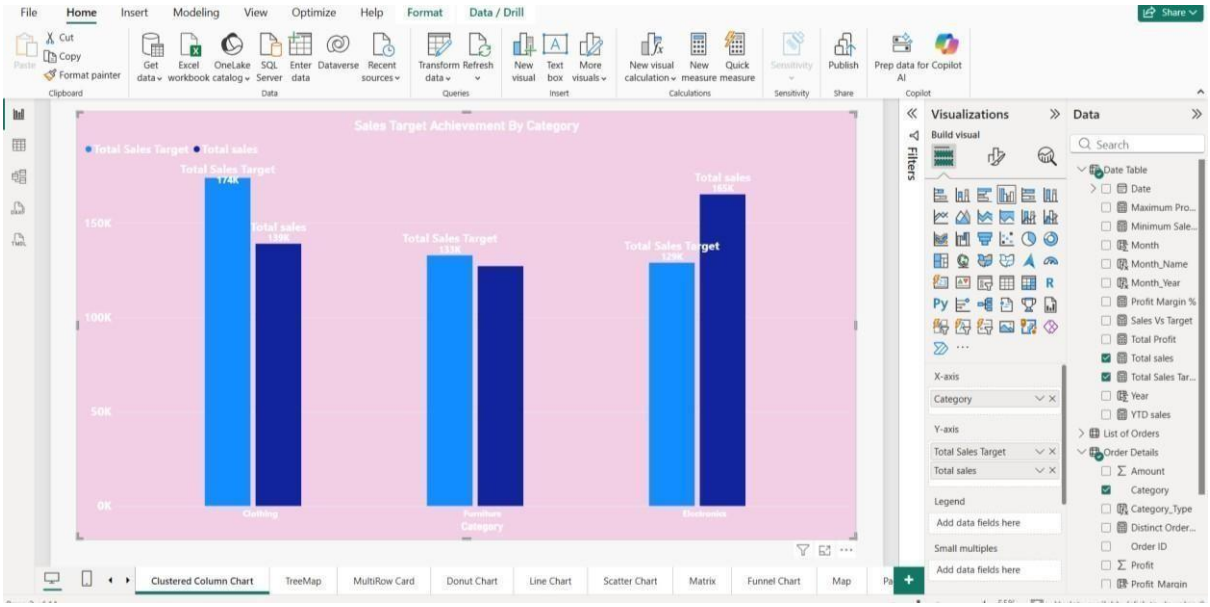
Cardinality = Many to One Cross Filter Directions = Single

DATA VISUALIZATION

1. **Sales Target Achievement by Category:** Compare actual sales with sales targets by category using a clustered column chart. Here . new measure TOTAL SALES TARGET is created.

TOTAL SALES TARGET = SUM('Sales target'[Target])





2. **Max Profit Margin by Sub-Category:** Analyze the maximum profit margin for each sub-category of products using a donut chart.

For this chart , the new column PROFIT_MARGIN is created in the ORDER DETAILS table.

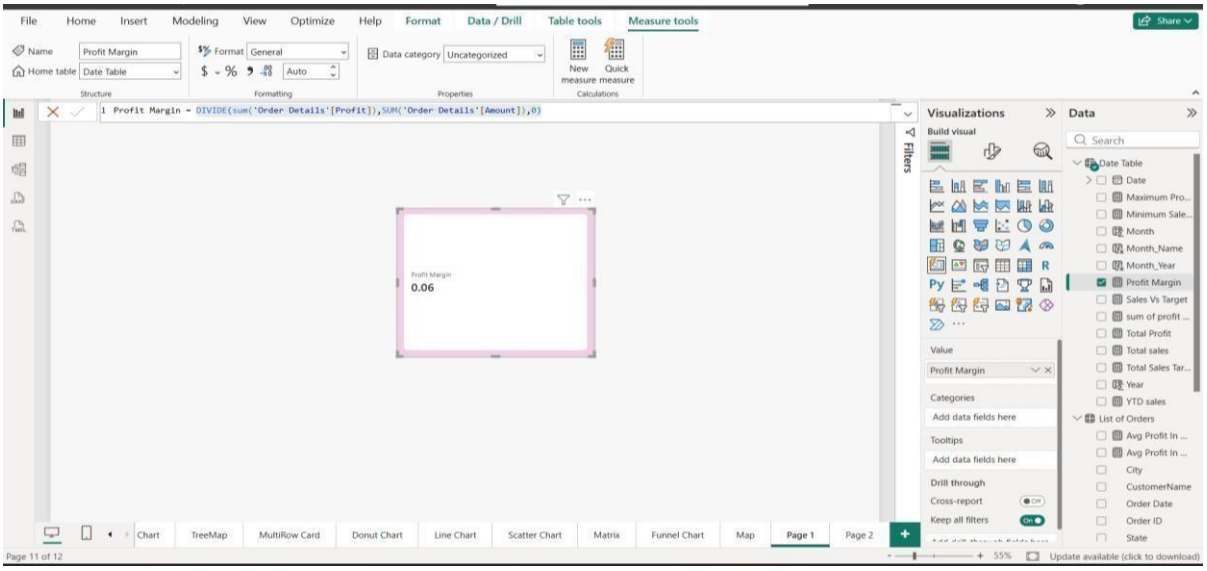
PROFIT_MARGIN = DIVIDE([Total Profit],[Total sales],0)

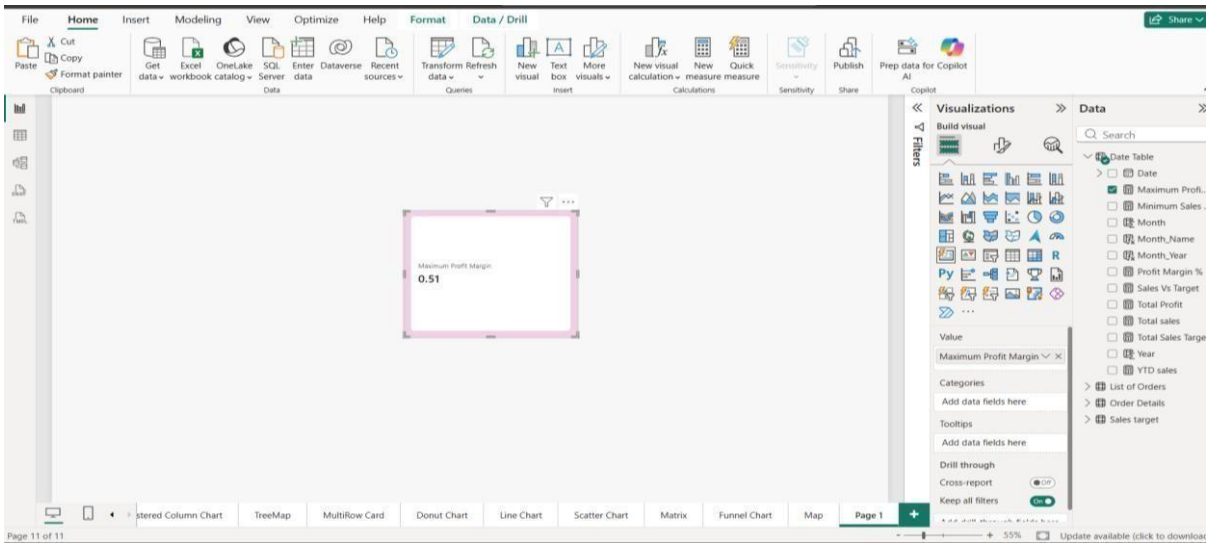
New 3 measures are created.

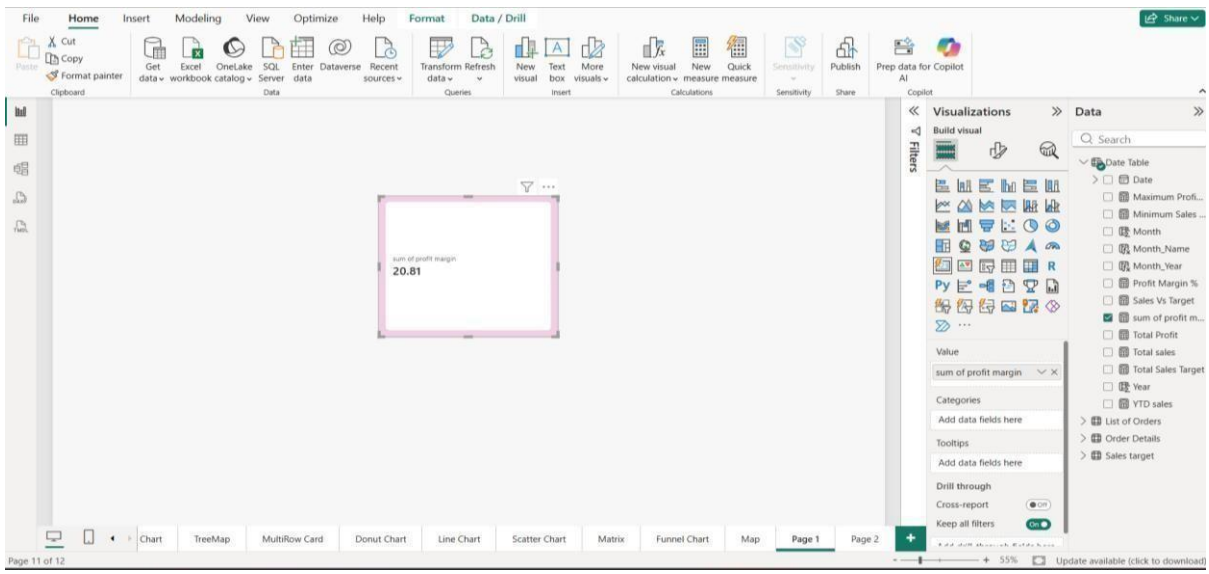
PROFIT MARGIN = DIVIDE (SUM ('Order Details'[Profit]), SUM ('Order Details'[Amount]), 0)

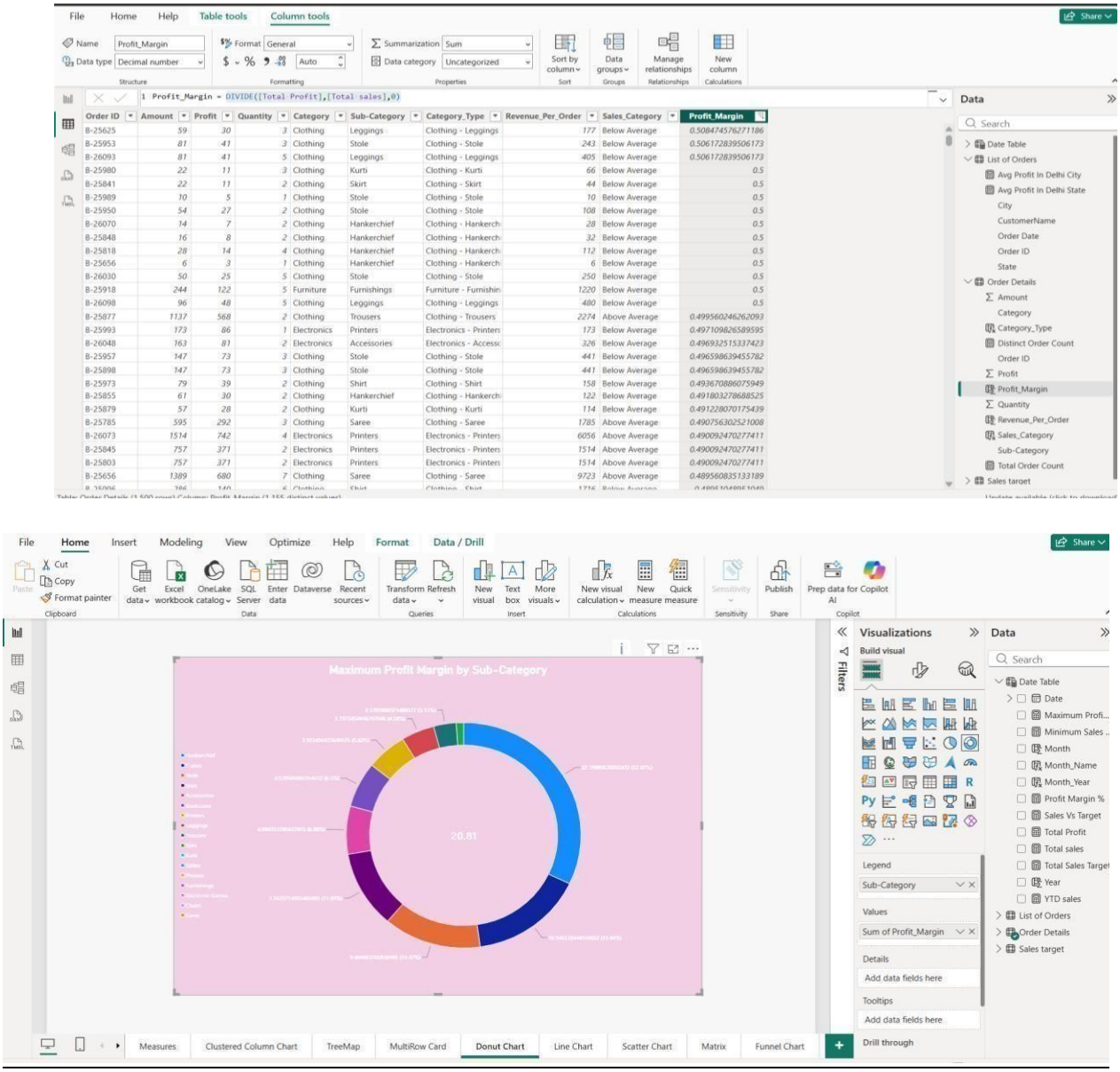
MAXIMUM PROFIT MARGIN = MAX('Order Details'[Profit_Margin]) SUM

OF PROFIT MARGIN = SUM('Order Details'[Profit_Margin])

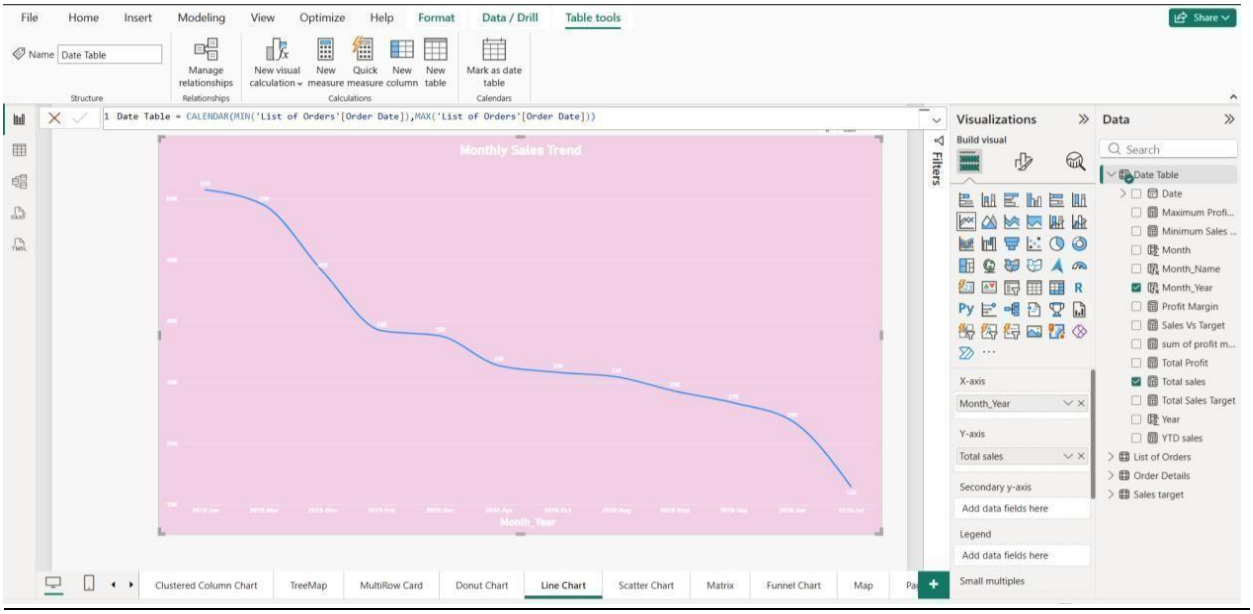

File Home Insert Modeling View Optimize Help Format Data / Drill Table tools Measure tools
Name Profit Margin
Home table Date Table
Format General
Data category Uncategorized
New measure
Quick measure
1 Profit Margin = DIVIDE(sum('Order Details'[Profit]),SUM('Order Details'[Amount]),0)
Profit Margin
0.06
Visualizations
Build visual
Data
Search
Value
Profit Margin
Categories
Add data fields here
Tooltips
Add data fields here
Drill through
Cross-report
Keep all filters
Chart TreeMap MultiRow Card Donut Chart Line Chart Scatter Chart Matrix Funnel Chart Map Page 1 Page 2
Page 11 of 12


File Home Insert Modeling View Optimize Help Format Data / Drill
Maximum Profit Margin
0.51
Visualizations
Build visual
Data
Search
Value
Maximum Profit Margin
Categories
Add data fields here
Tooltips
Add data fields here
Drill through
Cross-report
Keep all filters
Page 1
Page 11 of 11

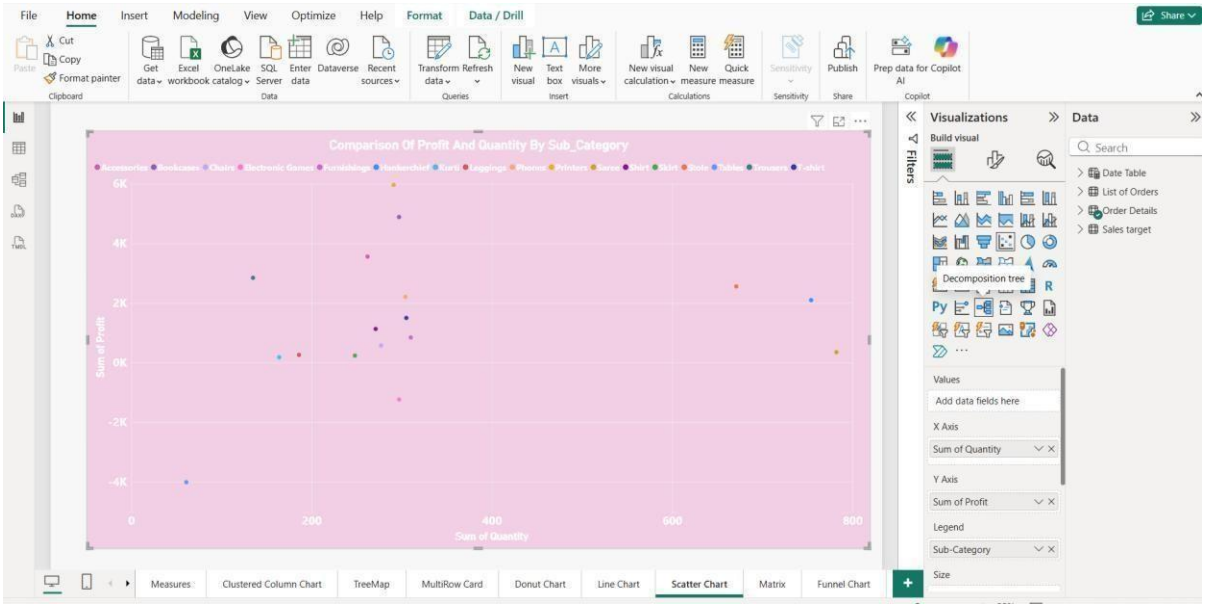

File Home Insert Modeling View Optimize Help Format Data / Drill
sum of profit margin
20.81
Visualizations
Build visual
Data
Search
Value
sum of profit margin
Categories
Add data fields here
Tooltips
Add data fields here
Drill through
Cross-report
Keep all filters
Chart TreeMap MultiRow Card Donut Chart Line Chart Scatter Chart Matrix Funnel Chart Map Page 1 Page 2
Page 11 of 12



3. **Monthly Sales Trend:** Show the trend of monthly sales over time using a line chart

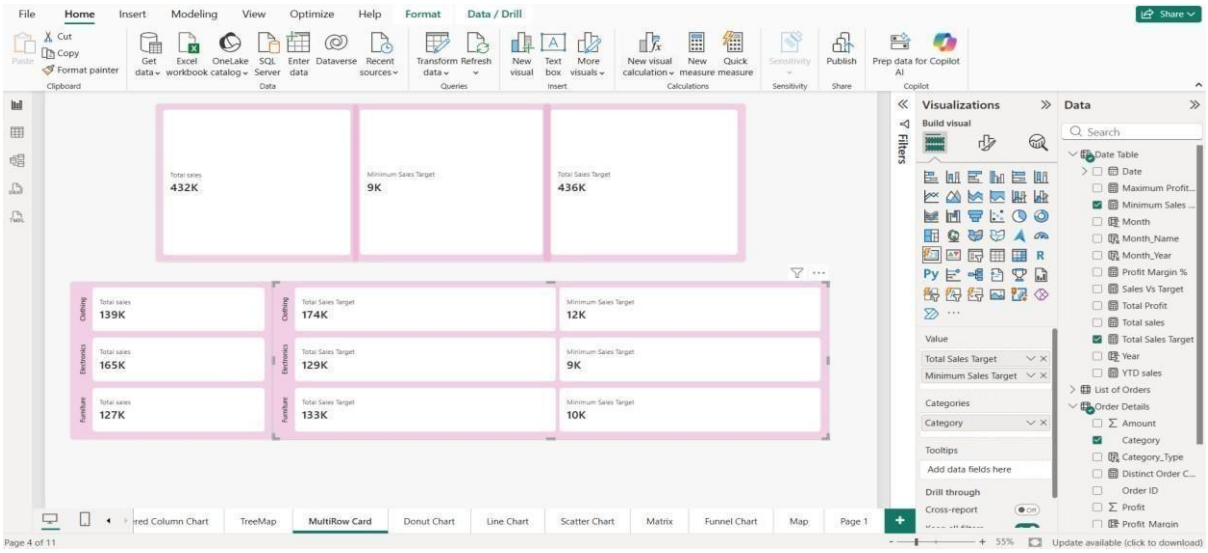


4. **Comparison of Profit and Quantity by Sub-Category:** Compare the relationship between profit and quantity sold for different subcategories using a scatter chart.



5. **Comparison of Total Sales Amount and Target:** Create cards to succinctly display the total sales amount alongside the sales target for quick comparison and analysis. Also, create a multi-row card to display the minimum target for each segment.

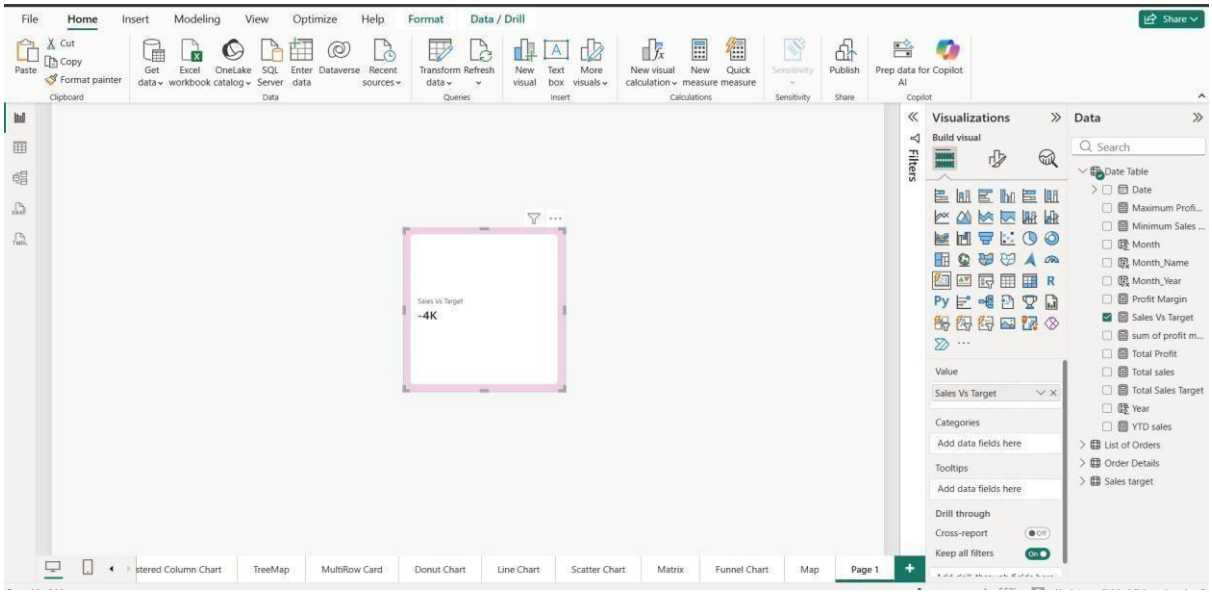
Total Sales Target = SUM('Sales target'[Target])
Minimum Sales Target = MIN('Sales target'[Target])

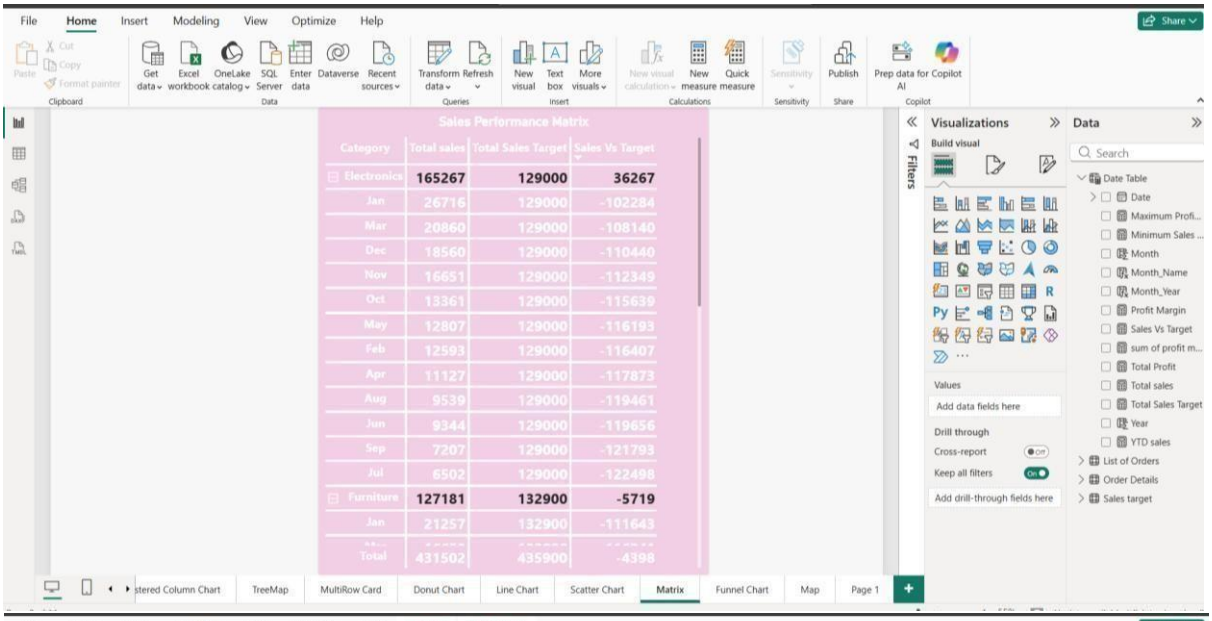


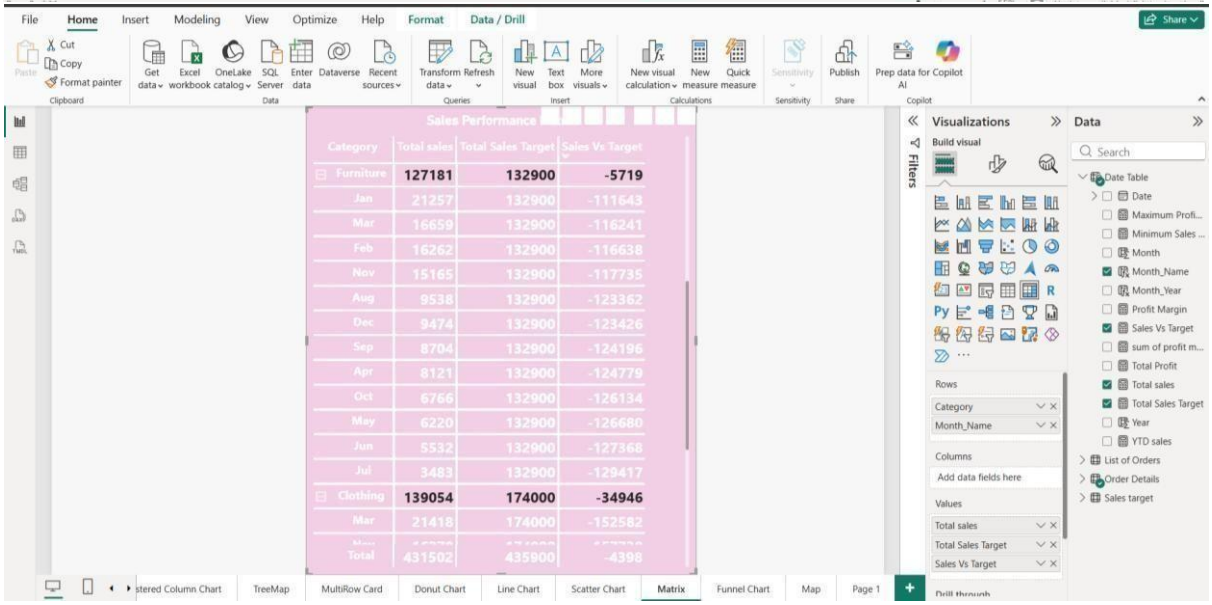
6. **Sales Performance Matrix:** Build a matrix view to analyze how actual sales compare to sales targets across different categories and months.

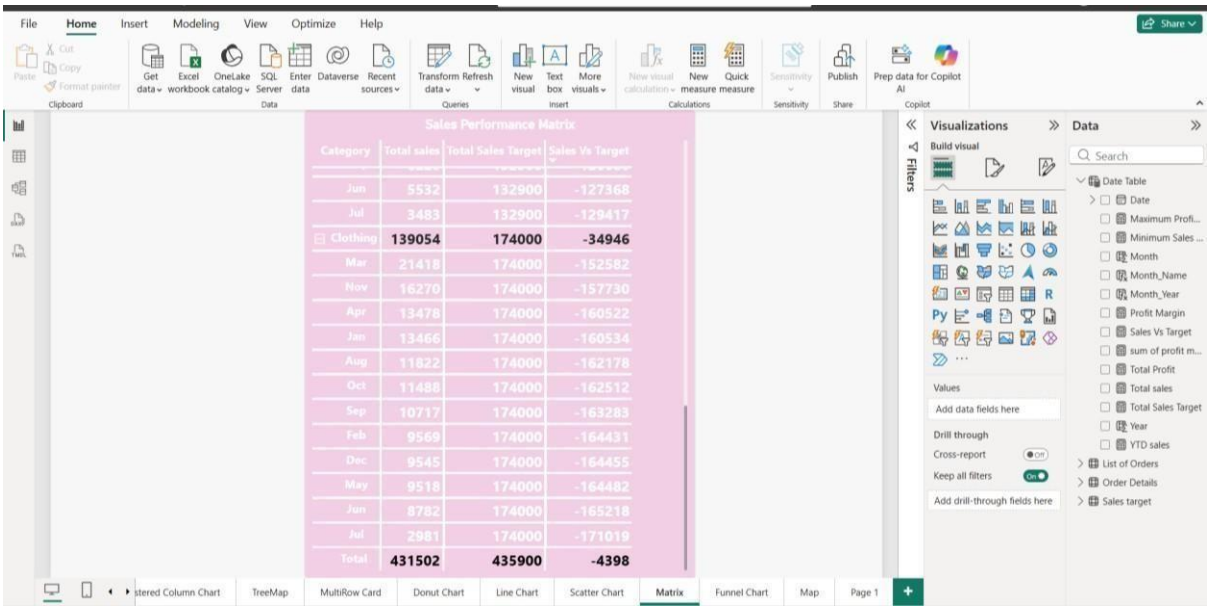
The new measure is created.

SALES VS TARGET = [Total sales] - [Total Sales Target]

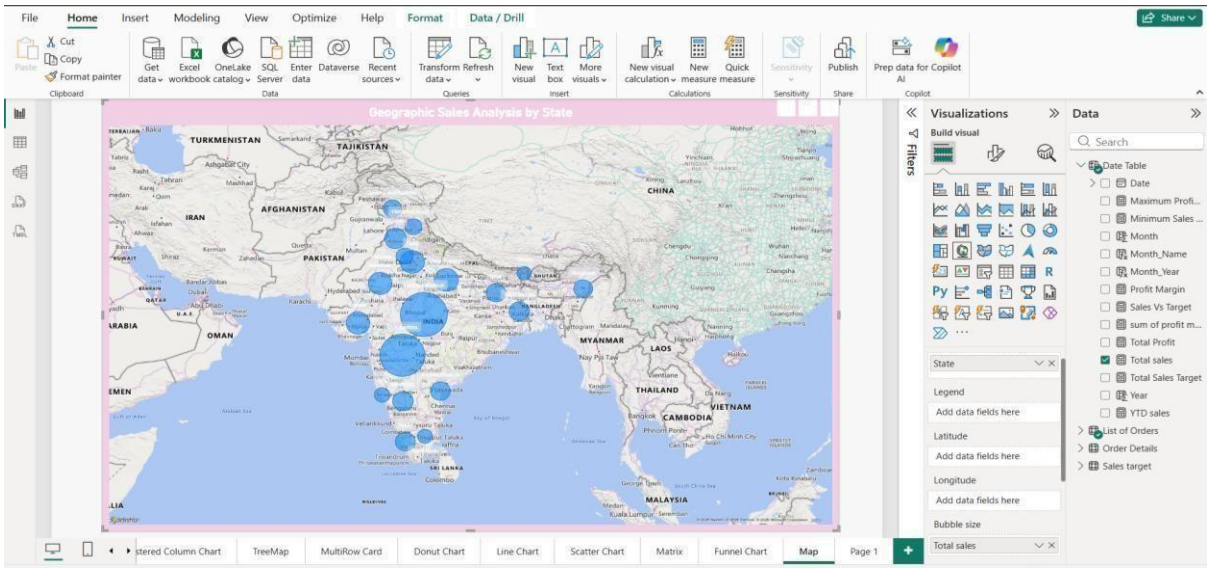

Sales Vs Target
-4K
Visualizations
Build visual
Data
Search
Value
Sales Vs Target
Categories
Add data fields here
Tooltips
Add data fields here
Drill through
Cross-report
Keep all filters
Page 1


Sales Performance Matrix
Category | Total sales | Total Sales Target | Sales Vs Target
Electronics 165267 129000 36267
Jan 26716 129000 -102284
Mar 20860 129000 -108140
Dec 18560 129000 -110440
Nov 16651 129000 -112349
Oct 13361 129000 -115639
May 12807 129000 -116193
Feb 12593 129000 -116407
Apr 11127 129000 -117873
Aug 9539 129000 -119461
Jun 9344 129000 -119656
Sep 7207 129000 -121793
Jul 6502 129000 -122498
Furniture 127181 132900 -5719
Jan 21257 132900 -111643
Total 431502 435900 -4398
Visualizations
Build visual
Data
Search
Values
Add data fields here
Drill through
Cross-report
Keep all filters
Add drill-through fields here
Page 1

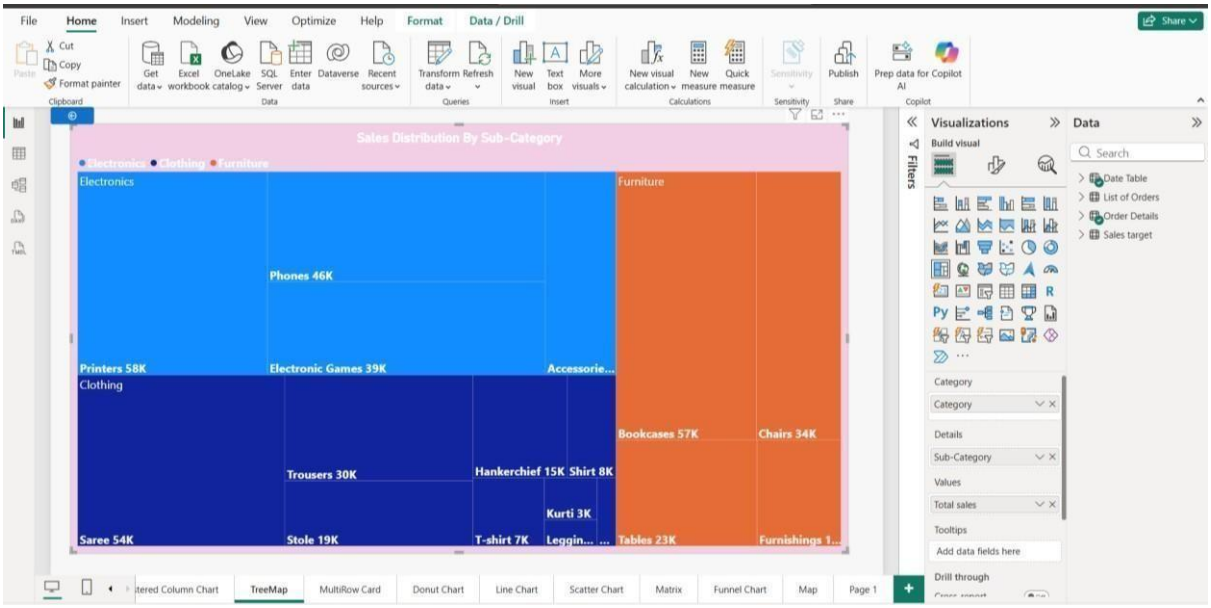

Sales Performance
Category | Total sales | Total Sales Target | Sales Vs Target
Furniture 127181 132900 -5719
Jan 21257 132900 -111643
Mar 16659 132900 -116241
Feb 16262 132900 -116638
Nov 15165 132900 -117735
Aug 9538 132900 -123362
Dec 9474 132900 -123426
Sep 8704 132900 -124196
Apr 8121 132900 -124779
Oct 6766 132900 -126134
May 6220 132900 -126680
Jun 5532 132900 -127368
Jul 3483 132900 -129417
Clothing 139054 174000 -34946
Mar 21418 174000 -152582
Total 431502 435900 -4398
Visualizations
Build visual
Data
Search
Rows
Category
Month_Name
Columns
Add data fields here
Values
Total sales
Total Sales Target
Sales Vs Target
Page 1



7. **Geographic Sales Analysis:** Visualize total sales on a map by city to identify regional sales patterns.



8. **Sales Distribution by Sub-Category:** Represent the sales distribution across different sub-categories using a Treemap.



9. **Order Count Analysis by State:** Create a funnel chart to visualize the distribution of order counts across different states.

